

Bypassing a False Shape Bridge: A Lean-Certified Direct Mediator Constructor in the Branch-C Farey Program

How a direct construction proves the target without rehabilitating the refuted classifier

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July 28, 2026

Abstract

A proof program may reach a correct target through an invalid intermediate classifier. The correct response is not to erase the counterexample to the classifier. It is to preserve the failed route and prove the target through a different dependency chain. Paper 10 did the first job inside the finite Branch-C Farey mediator program. It machine-checked that the canonical pair $N=3$, $a=(0, 1)$, $b=(2, 3)$ satisfies the determinant premise yet lies outside the then-current union of endpoint shapes. This companion performs the second job. It records the later direct-constructor route that proves the finite arbitrary-endpoint listed-mediator target without using the false bridge.

The route consists of 19 copied Lean sources from BC-054IW through BC-055O and 18 packet claim signatures. BC-055N is retained as a disclosed formal-only intermediate rather than being given a fabricated packet. Restricted constructors and exact finite checkers lead through mediant, overflow, unit-certificate, modular-interval, determinant-interval, and modular-inverse reductions. BC-055N reduces the remaining task to a Euclidean algebra step. BC-055O proves that step and closes the left-endpoint bounded inverse, the direct certified-mediator constructor premise, and the arbitrary-endpoint listed-mediator target in the copied finite Farey formalization.

The logical relation is bypass, not repair. The later theorem does not put the Paper 10 counterexample inside the old shape union. The two results can therefore be true together: the old coverage bridge is false, while the listed-mediator target is true through another route. The package pins Lean 4.30.0 and Mathlib, verifies every governed hash, compiles all 19 sources, and reports the axioms of the promoted endpoint theorems. The result remains finite and formal. It proves no finite C-22, infinite transfer, HSIT, SIT / Riemann bridge, Riemann Hypothesis, full or physical Super Information Theory, or empirical claim.

Scope and evidence boundary

selfawarenetworks.com Final-preprint 1.0 - July 28, 2026

Companion to Paper 10: [doi:10.5281/zenodo.21231896](https://doi.org/10.5281/zenodo.21231896) Mathematical development assistance: OpenAI Codex. The author determined the research direction, claim boundary, and final text.

1. The problem before the terminology

Suppose a proof needs to show that every admissible object has a desired witness. One attempted route first classifies every object into a list of shapes and then supplies a witness for each shape. Let $P(x)$ mean that x is admissible, $S(x)$ mean that x belongs to the proposed shape union, and $T(x)$ mean that x has the desired witness. The classification bridge is

$$B_{\text{shape}} : \quad \forall x, P(x) \rightarrow S(x). \quad (1)$$

The actual target is

$$T_{\text{witness}} : \quad \forall x, P(x) \rightarrow T(x). \quad (2)$$

Proving each listed shape sufficient does not prove Equation (1). A single admissible object outside the shape union refutes the bridge. That refutation does not settle Equation (2), because another construction may produce the witness without classifying the object:

$$\neg B_{\text{shape}} \not\Rightarrow \neg T_{\text{witness}}. \quad (3)$$

This is the exact relationship between the two Branch-C papers. Paper 10 proves the left statement below. This companion proves the right statement through a different route:

$$\neg B_{\text{shape}} \quad \text{and} \quad T_{\text{mediator}}. \quad (4)$$

Only after that distinction is clear do the local theorem names become useful. Paper 10 refutes `arbitrary EndpointShapeBridgeContract`. The later `BC-054IW-through-BC-055O` route proves `arbitrary EndpointListedMediatorTargetContract` through a direct constructor. The second theorem bypasses the first proposition; it does not make that proposition true.

2. Established Farey setting and local formalization

Farey sequences contain reduced rational numbers with bounded denominators in increasing order [1]. The copied Lean development represents a rational-like endpoint by a pair of natural numbers,

$$\text{RawPair} := \mathbb{N} \times \mathbb{N}, \quad a = (p, q). \quad (5)$$

For a finite order N , the canonical list contains pairs satisfying

$$1 \leq q \leq N, \quad p \leq q, \quad \gcd(p, q) = 1. \quad (6)$$

For $a = (p_a, q_a)$ and $b = (p_b, q_b)$, strict Farey order is represented by cross multiplication,

$$a \prec b \iff p_a q_b < p_b q_a, \quad (7)$$

and the local determinant is

$$\Delta(a, b) := p_b q_a - p_a q_b. \quad (8)$$

The endpoint premise used by the constructor route is therefore

$$C_N(a, b) := a, b \in \text{rawPairsCanonical}(N) \wedge a \prec b \wedge 1 < \Delta(a, b). \quad (9)$$

A listed mediator is a canonical pair m at the same finite order, strictly between the endpoints:

$$M_N(a, m, b) := m \in \text{rawPairsCanonical}(N) \wedge a \prec m \wedge m \prec b. \quad (10)$$

The target is the finite existence statement

$$T_{\text{mediator}} : \quad \forall N \, a \, b, \, C_N(a, b) \rightarrow \exists m, \, M_N(a, m, b). \quad (11)$$

The word “arbitrary” in the local theorem name ranges over the finite endpoint objects defined by Equations (5)-(11). It does not denote an infinite operator theorem, arbitrary physical systems, or every external use of the word *mediator*.

Three exact contract names control the endpoint of the formal route:

Contract	Role	Companion status
<code>leftEndpointBoundedInverseContract</code>	Supplies bounded modular inverse data for a reduced proper left endpoint.	Proved by BC-055O.
<code>directArbitraryCertifiedMediatorConstructorPremise</code>	Supplies a certified mediator constructor for arbitrary canonical determinant-greater-than-one pairs.	Proved by BC-055O through the accumulated reductions.
<code>arbitraryEndpointListedMediatorTargetContract</code>	States Equation (11) in the copied formalization.	Proved by the existing constructor-to-target reduction once the direct premise closes.

3. Route inventory

The companion preserves every formal stage from BC-054IW through BC-055O. Each row describes what the stage contributes and what it still leaves open at that point.

Stage	Contribution	Remaining boundary at that stage
BC-054IW	Proves the top-endpoint mediator (n-1, n) without the earlier successor-coprime side condition and records constructor / listed-target equivalence.	Does not prove arbitrary constructor existence.
BC-054IX	Proves the ordinary mediant is a listed strict mediator when the denominator sum is within order and the mediant is reduced.	Leaves overflow and non-reduced cases.
BC-054IY	Proves a bounded reduced scaling of the mediant works under an explicit positive-scale certificate.	Does not prove the certificate exists for every pair.
BC-054IZ	Equates the direct constructor premise with finite Boolean witness-search closure over canonical raw pairs.	Does not prove the all-order Boolean result.
BC-055A	Implements a sound fixed-order checker and proves closure for $N=0 \dots 12$.	Finite computation is evidence / theorem at those orders only.
BC-055B	Proves exact checker equivalence at fixed N and connects all-order search closure to the direct premise.	Does not prove every checker returns true.
BC-055C	Recasts the remaining issue as a Farey-neighbor determinant-greater-than-one obstruction contract.	Does not prove the obstruction theorem.
BC-055D	Closes the bounded-mediator case and isolates denominator overflow.	Leaves the overflow-only case.
BC-055E	Closes reducible overflow mediants and isolates primitive overflow.	Leaves primitive-overflow witness existence.

Stage	Contribution	Remaining boundary at that stage
BC-055F	Shows left / right unit-determinant interior certificates provide listed strict mediators.	Does not produce those certificates universally.
BC-055G	Handles primitive-overflow boundary denominators and reduces to the interior unit-certificate contract.	Leaves the interior existence theorem.
BC-055H	Reduces interior unit certificates to bounded modular-interval witnesses.	Leaves modular witness existence.
BC-055I	Reduces modular witnesses to determinant-interval witnesses.	Leaves determinant witness existence.
BC-055J	Reduces determinant witnesses to modular-inverse threshold witnesses.	Leaves threshold witness existence.
BC-055K	Shows paired bounded inverse data imply the threshold dichotomy and direct premise.	Leaves paired inverse data existence.
BC-055L	Assembles paired inverse data from standalone left / right endpoint inverse contracts.	Leaves both endpoint inverse contracts.
BC-055M	Derives the right endpoint inverse contract from the left endpoint contract by complement arithmetic.	Leaves the left endpoint inverse theorem.
BC-055N	Formal-only intermediate: reduces the left inverse theorem to a Euclidean algebra step via <code>Nat.gcd.induction</code> .	Leaves the algebra step itself.
BC-055O	Proves the Euclidean product / complement identities, closes the algebra step, left inverse, direct premise, and listed target.	Higher Branch-C, infinite, RH, SIT, and physical targets remain open.

The table is a dependency narrative, not a claim that every intermediate packet is logically imported as a separate module by the final file. The sources are cumulative formal snapshots. The exact final theorem types and their copied source are authoritative.

3.1 Cumulative snapshots and evidential roles

The route files are preserved as dated cumulative snapshots rather than as a newly refactored Lean library. That choice has two consequences. First, a later file can contain definitions and theorems first introduced in an earlier stage without importing that earlier file as a module. The manuscript therefore uses the stage sequence to describe development and dependency of ideas, while the replay treats each source file as an independently compiled artifact. Second, the packet signatures and Lean files answer different questions. A packet signature records the intended claim boundary and source provenance; a Lean file records the term accepted by the checker. Neither is silently substituted for the other.

BC-055N makes that distinction visible. It is required to understand the final reduction, but no standalone packet signature was found for it. The package does not fabricate one. It labels BC-055N formal-only, includes its exact bytes, and requires the replay inventory to report one and only one formal-only stage. This preserves a documented absence instead of turning an archival gap into an apparently complete provenance claim.

The fixed-order computations at BC-055A have a similarly bounded role. Their closure for $N=0 \dots 12$ is an exact result for those inputs and a useful check on the definitions. It is not induction over all orders and is not used as a numerical extrapolation. The later Euclidean proof route, not the finite sample, is what closes the formal arbitrary-endpoint target.

4. The Euclidean endpoint

The remaining left-endpoint problem can be written as follows. For coprime natural numbers p, q with $1 < q$ and $p < q$, find natural numbers x, u such that

$$\begin{aligned} 1 &\leq x \leq q, \\ px + 1 &= uq. \end{aligned} \tag{12}$$

This is bounded inverse data with the sign convention used by the constructor route. BC-055N shows how a Euclidean step would lift inverse data for $(n \% m, m)$ to inverse data for (m, n) and then uses `Nat.gcd.induction` to reduce the general theorem to the algebra step.

BC-055O proves the needed identities. If the recursive witness satisfies

$$(n \bmod m)z + 1 = tm, \tag{13}$$

then the quotient / remainder identity lifts it:

$$m \left(\left\lfloor \frac{n}{m} \right\rfloor z + t \right) = nz + 1. \tag{14}$$

Writing $M = \lfloor n/m \rfloor z + t$, the complementary branch uses

$$m(n - M) + 1 = (m - z)n \tag{15}$$

under the formal boundedness hypotheses. The result closes `leftEndpointBoundedInverseContract`. BC-055M then supplies the right endpoint contract; BC-055L and the preceding reductions supply the direct constructor; the existing reduction supplies the listed target.

The paper does not present these displayed identities as a new theorem of elementary number theory detached from the formal definitions. Their relevance is their exact role in the copied Lean contract chain.

5. Why this is a bypass

Repairing the old bridge would require changing its domain, expanding the shape taxonomy, or proving that the original union actually covered the counterexample. None occurred. The proposition refuted by BC-054IS stays refuted.

The direct constructor route changes proof architecture. It asks for bounded inverse data and turns that data into a mediator certificate. A direct proof of the target does not imply that an earlier taxonomy-based lemma was true.

This gives a useful formal-workflow lesson:

1. Keep the counterexample as a first-class theorem artifact.
2. Do not weaken its wording after a different route succeeds.
3. State the new target and dependency chain independently.
4. Record whether the new route repairs, restricts, replaces, or bypasses the old route.
5. Preserve both artifacts and link them through metadata rather than silently rewriting history.

That workflow is the principal reason for a companion paper rather than a new version of Paper 10.

5.1 Why a companion is the conservative publication unit

A replacement version of Paper 10 would create an avoidable ambiguity. Readers could reasonably ask whether the counterexample had been withdrawn, whether the old contract had changed, or whether the later constructor retroactively made the refuted lemma true. A separate companion keeps the proposition, date, and proof object of the negative result stable while giving the positive route its own claim register and review history.

The relationship is also asymmetric. Paper 10 is sufficient to show that the old shape bridge fails, without depending on this companion. The companion needs Paper 10 only to explain which tempting inference is prohibited; its positive theorem route is the BC-054IW-through-BC-055O constructor chain. Accordingly, future metadata should express supplementation or relation, not replacement, correction, or retraction. Provider terminology must be checked at release time because repository relation vocabularies can differ.

This separation improves adversarial review. A reviewer can challenge the counterexample, the direct constructor, the prose description of their logical relation, or the proposed metadata independently. Failure in one review lane does not silently rewrite the evidence in another. It also prevents a later higher-level Branch-C claim from being smuggled into either paper merely because both use the same finite Farey vocabulary.

6. Reproducibility package

The package contains:

- 19 copied .lean files, BC-054IW through BC-055O;
- 18 copied packet claim signatures;
- a disclosure that BC-055N is formal-only;
- leanprover/lean4:v4.30.0 and Lean commit d024af099ca4bf2c86f649261ebf59565dc8c622;
- Mathlib revision c5ea00351c28e24afc9f0f84379aa41082b1188f;
- exact Lake and dependency manifests;
- a generated source / hash manifest;
- a replay wrapper that scans for `sorry`, `admit`, `custom axiom`, and `unsafe`, compiles every copied file, and prints axioms for the promoted BC-055O endpoint theorems.

Lean provides the kernel-checked theorem environment [2], while Mathlib supplies the pinned formal library used by these sources [3]. Compilation is evidence that Lean accepted the copied theorem terms under that environment. It is not a substitute for mathematical review of whether the definitions are the intended Farey objects or whether a local theorem has a higher-level interpretation.

The axiom report is also part of the trust boundary. Classical choice, propositional extensionality, quotient soundness, and any native-decision dependencies must be reported rather than hidden. The final decision theorem may have a different axiom profile from supporting theorems; this distinction is preserved in the replay output.

6.1 What replay can and cannot establish

The reproducibility claim has three layers. Byte integrity checks that the copied pins, signatures, and Lean sources match the generated proof manifest. Policy scanning checks that the copied Lean sources contain no `sorry`, `admit`, `custom axiom`, or `unsafe` declaration. Pinned compilation checks that all 19 files are accepted by Lean 4.30.0 in the recorded Mathlib environment, and the final report asks Lean to print the axioms of each promoted BC-055O theorem.

Those checks establish a reproducible local formal result, but they do not establish semantic adequacy by themselves. The definitions of canonical pair, listed mediator, strict order, bounded inverse, and the constructor-to-target interface remain the objects a mathematical reviewer should inspect. In particular, successful compilation does not license replacing the phrase “in the copied finite Farey formalization” with an unqualified physical or infinite-operator claim.

The route contains 19 copied Lean files and 18 packet claim signatures; BC-055N is formal-only. That count is enforced in both the proof manifest and the package validator so that omission, duplication, or a later undocumented packet cannot pass unnoticed.

7. Claim boundary

BC-055O proves only its exact finite formal contracts. No finite C-22, infinite operator transfer, zeta-zero spectral identification, HSIT, SIT / Riemann bridge, Riemann Hypothesis, full SIT, physical SIT, or empirical validation follows.

The result also does not establish that the formalized target has the same importance as a classical named theorem outside this program. Publication language must lead with the local definitions and exact theorem names. Terms such as “arbitrary endpoint” and “closure” must not be detached from the finite model and made to sound like infinite analytic closure.

8. Companion relation and release architecture

Paper 10 remains citable under DOI 10.5281/zenodo.21231896 [4]. This companion requires a separate record because it proves a different proposition. Its provider metadata should identify Paper 10 as the related counterexample paper and use the provider’s current supplement or related-work relation without claiming replacement, correction, or retraction.

The public bundle contains the manuscript, searchable PDF, copied formal sources, environment pins, proof manifest, replay result, and a package-wide SHA-256 manifest. Zenodo and OSF should receive the same governed scientific files and relation statement. The upload is scheduled with the complete paper-portfolio release tranche.

9. Limitations and discriminating review

The theorem’s strongest limitation is semantic scope. The Lean kernel can certify that the theorem term inhabits the stated local type. It does not determine whether a reader intends a different definition of Farey adjacency, mediator, or endpoint domain. The manuscript therefore prints the finite definitions and refuses every infinite, spectral, physical, or empirical promotion.

The cumulative snapshots are another limitation. They preserve development history and exact bytes, but they are less modular than a refactored library. A useful independent review should therefore test three separable propositions:

1. the local definitions match Equations (5)-(11);
2. the 19-file route culminates in the three stated BC-055O contracts;
3. the prose never promotes those contracts beyond their finite domain.

Failure of the first or second proposition would challenge the mathematical result. Failure of the third would require a wording correction. None would make the Paper 10 shape bridge true.

10. Claim register

Claim	Status
Paper 10 refutes one endpoint-shape bridge.	Retained exact finite theorem.
The old bridge is repaired.	False / prohibited.
The constructor route bypasses the old bridge.	Retained logical relation.
BC-055N is a packet-backed result.	False; it is formal-only.
BC-055O proves the left bounded inverse, direct constructor premise, and listed target in the copied formalization.	Retained exact finite theorem scope.
The companion proves finite C-22, HSIT, SIT / Riemann, RH, full SIT, or physics.	False / prohibited.
The package compiles 19 copied files under pinned Lean / Mathlib.	Reproducibility claim, contingent on generated replay result.
This companion replaces or retracts Paper 10.	False / prohibited.
A separate final-preprint package is available.	Prepared with manuscript, PDF, formal sources, pins, replay evidence, metadata, and hashes.

References

1. A. O. Matveev, *Farey Sequences: Duality and Maps Between Subsequences*, De Gruyter, 2017. DOI: 10.1515/9783110547665.
2. L. de Moura and S. Ullrich, “The Lean 4 Theorem Prover and Programming Language,” CADE-28, 2021. DOI: 10.1007/978-3-030-79876-5_37.
3. The Mathlib Community, “The Lean Mathematical Library,” CPP 2020. DOI: 10.1145/3372885.3373824.
4. M. Blumberg, “A Lean-Certified Endpoint-Shape Counterexample in the Branch-C Farey Mediator Program,” Zenodo, 2026. DOI: 10.5281/zenodo.21231896.

Conclusion

The formal record contains both a failure and a success, and neither should be erased. BC-054IS proves that one endpoint-shape bridge is false. BC-055O proves the listed-mediator target through a different bounded-inverse constructor route. The success bypasses the failure; it does not convert the false bridge into a true statement.

The companion’s defensible contribution is the exact route, pinned replay, and logical distinction. Everything above the finite Farey theorem types remains outside scope.